Associating data with Pandas: joins

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The introductory chapter to `Pandas` presented how data were organized as *DataFrames* and how the `Pandas` ecosystem can be useful to perform simple operations on datasets. This chapter consolidates these principles by introducing one of the most common operations in data science: associating data through a join.

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Skills to be acquired by the end of this chapter

- Retrieve an official dataset from Insee;
- Understand the logic of a join and know how to choose the appropriate pivot;
- Merge several `DataFrames` based on common characteristics;
- Know the main identifiers used in French data.

The hands-on exercises in this tutorial come rather late, once the principles of joins and identifiers have been covered: see Tip 1 and Tip 2.

1 Introduction

The introductory chapter to `Pandas` presented the concept of data organized in the form of a *DataFrame* and the practicality of the `Pandas` ecosystem for performing simple operations on a dataset.

One of the benefits of modern data science tools, especially `Pandas`, is the ease with which they allow restructuring sources to work on multiple datasets in a project. This chapter consolidates the principles previously seen by exploring one of the most frequent operations in the data scientist's toolbox: associating data through common characteristics.

It is rare to work exclusively on a raw source. A dataset generally gains value when compared to other sources. For researchers, this allows contextualizing the information present in one source by comparing or associating it with other sources. For data scientists in the private sector, it often involves linking information about the same person in multiple customer databases or comparing customers with each other.

Performing this work simply, reliably, and efficiently is essential for data scientists as this task is common. Fortunately, `Pandas` handles this very well with structured data. In the following chapters, and also throughout the section on text data processing, we will see how to handle less structured data.

Thanks to this enrichment work, we will be able, in the next chapter, to deepen our understanding of a real world phenomenon through detailed descriptive statistics. This is an essential step before moving on to inferential statistics, the approach that consists of formalizing and generalizing correlations or causal relationships between observed characteristics and a phenomenon.

1.1 Environment

The previous chapter used almost exclusively the `Pandas` library since we had a ready-to-use source. In this chapter, we will need to fetch data that is a bit less accessible: this will require a little more work.

The general principles of data retrieval will be presented in the chapter dedicated to APIs. We will give the data retrieval code directly in this chapter so that we can focus on the core of our problem: associating these different datasets with one another.

The essential packages to start this chapter are as follows:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import requests
```

To obtain reproducible results, you can set the seed of the pseudo-random number generator.

```
import numpy as np
np.random.seed(123)
```

1.2 Data used

This tutorial continues the exploration of the dataset from the previous chapter:

- Greenhouse gas emissions estimated at the municipal level by ADEME. The dataset is available on data.gouv and can be directly queried in Python with this URL;

The issues of data enrichment (associating one source with another based on common characteristics) will be presented using two sources produced by Insee:

- The official geographic code, a database of unique French zip codes;
- The *Filosofi* data, a source on French income at a fine spatial scale constructed by Insee from tax returns and social benefit information. In this case, we will use, at the municipal level, median income data and total population[1]. This will help us compute new emissions indicators and interpret them.

To import this data, we will directly query the `Melodi` API, Insee's new entry point for disseminating its main data sources, including *Filosofi*. This is no longer a dedicated `Python` package but a standard API, queried with `requests`. More details on handling JSON with Python will be given in the chapter dedicated to APIs.

> **i** Retrieving open data produced by Insee
>
> A previous version of this tutorial relied on the community *package* `pynsee` , which simplifies access to Insee's official data.
>
> However, the sources useful to us here have been integrated into a new Insee API named `Melodi`. As this is an official product of Insee's dissemination offer, we will favor it. The downside is that we do not have, as with `pynsee` , a ready-to-use *dataframe*. The data preparation code will therefore be a bit more complex.
>
> The API returns structured `JSON` data: each observation associates dimensions (for example `GEO` for the geographic area or `TIME_PERIOD` for the period) with a value (`OBS_VALUE_NIVEAU`). More details on handling JSON with Python will be given in the chapter dedicated to APIs.
>
> Each source has a unique identifier (for example `DS_FILOSOFI_CC` for municipal Filosofi data, `DS_POPULATIONS_HISTORIQUES` for legal populations). The main entry point has the form `https://api.insee.fr/melodi/data/{identifier}`, which can be filtered by dimension (geographic area, period, indicator...). The structure of each file depends on the source requested, so some source-specific work is generally needed.
>
> Exploring the structure of a source To know the structure of a source (dimensions, available indicators...), you can look at its metadata, also available in `JSON` format:
>
> ```
> import requests
> requests.get("https://api.insee.fr/melodi/catalog/DS_FILOSOFI_CC").json()
> ```
>
> The list of sources available on the Melodi API can be browsed on Insee's API portal.

2 Retrieving data for this chapter

2.1 French carbon emissions dataset

As explained in the previous chapter, these data can be imported very simply with `Pandas`:

```
import pandas as pd

url = "https://data.ademe.fr/data-fair/api/v1/datasets/igt-pouvoir-de-rechauffement-global/convert"
```

[1] For demographic data, we directly use the legal populations published by Insee, derived from the census, rather than a fiscal-population proxy. An open exercise is also proposed to construct population aggregates from anonymized individual census data (the detailed files).

```
emissions = pd.read_csv(url)
emissions.head(2)
```

	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire
0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172
1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207

We will already keep the names of the emitting sectors present in the database to simplify subsequent uses:

```
secteurs = emissions.select_dtypes(include='number').columns
secteurs
```

	columns
0	Agriculture
1	Autres transports
2	Autres transports international
3	CO2 biomasse hors-total
4	Déchets
5	Energie
6	Industrie hors-énergie
7	Résidentiel
8	Routier
9	Tertiaire

Subsequent exploitations of these data will use the departmental dimension, the construction of which we demonstrated in the previous chapter:

```
emissions['dep'] = emissions["INSEE commune"].str[:2]
```

We will use the Filosofi data (income data) at the municipal level, in its most recent vintage (2023 at the time of writing). This is not the same year as the Ademe emissions data (regularly updated, it now covers 2025), so it is not perfectly rigorous, but it will still illustrate the main functionalities of `Pandas`.

At the municipal level, the Melodi source `DS_FILOSOFI_CC` only disseminates the median standard of living (`MED_SL`) and the poverty rate (`PR_MD60`); the other indicators (deciles, Gini index, breakdown of disposable income…) are only available at the département, region and metropolitan France levels. We add the municipal population, available from the Melodi source `DS_POPULATIONS_HISTORIQUES`, as well as municipality names, taken from the official geographic code.

The code to retrieve this data is as follows:

```
import requests
import pandas as pd

def get_melodi(dataset, dimension, code, value_name, time_period, geo_level="COM"):

  url = f"https://api.insee.fr/melodi/data/{dataset}"

  params = {
    "GEO": geo_level,
    dimension: code,
    "TIME_PERIOD": time_period,
```

```
    "maxResult": 40000,
  }

  response = requests.get(url, params=params, timeout=60)
  response.raise_for_status()
  observations = response.json()["observations"]

  return pd.DataFrame(
    {
      "CODGEO": obs["dimensions"]["GEO"].split("-")[-1],
      value_name: obs["measures"]["OBS_VALUE_NIVEAU"].get("value"),
    }
    for obs in observations
  )

niveau_vie = get_melodi("DS_FILOSOFI_CC", "FILOSOFI_MEASURE", "MED_SL", "NIVVIE_MEDIAN", 2023)
taux_pauvrete = get_melodi("DS_FILOSOFI_CC", "FILOSOFI_MEASURE", "PR_MD60", "TAUX_PAUVRETE",
2023)
population = get_melodi("DS_POPULATIONS_HISTORIQUES", "POPREF_MEASURE", "PMUN", "POPULATION",
2023)

url_cog_2023 = "https://www.insee.fr/fr/statistiques/fichier/6800675/v_commune_2023.csv"
url_backup = "https://minio.lab.sspcloud.fr/lgaliana/data/python-ENSAE/cog_2023.csv"

try:
  cog_2023 = pd.read_csv(url_cog_2023)
except requests.exceptions.Timeout:
  cog_2023 = pd.read_csv(url_backup)

noms_communes = (
  cog_2023
  .loc[cog_2023["TYPECOM"] == "COM", ["COM", "LIBELLE"]]
  .rename(columns={"COM": "CODGEO", "LIBELLE": "LIBGEO"})
)

filosofi = (
  niveau_vie
  .merge(taux_pauvrete, on="CODGEO", how="outer")
  .merge(population, on="CODGEO", how="outer")
  .merge(noms_communes, on="CODGEO", how="left")
  [["CODGEO", "LIBGEO", "POPULATION", "NIVVIE_MEDIAN", "TAUX_PAUVRETE"]]
)

filosofi["POPULATION"] = filosofi["POPULATION"].astype(int)
filosofi["dep"] = filosofi["CODGEO"].str[:2]
```

The resulting `DataFrame` looks like this:

```
filosofi.sample(3)
```

	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE	dep
31453	80574	Mouflers	96	NaN	NaN	80
21028	57332	Hombourg-Haut	5976	21870.0	24.0	57
30870	79289	Saint-Pierre-des-Échaubrognes	1384	25800.0	NaN	79

`Pandas` automatically handles variable types. It does that quite well. If in doubt, you can always use the `dtypes` method to check.

A quick glance at the data gives a fairly precise idea of how the data are organized. We notice that some variables in `filosofi` seem to have many missing values (statistical secrecy), while others seem complete. If we want to exploit `filosofi`, we need to pay attention to the chosen variable.

Our immediate goal is therefore to link the information contained in our two datasets. Otherwise, we risk being frustrated: we will want to know more about carbon emissions but will be very limited in the possibilities of analysis without adding additional information from `filosofi`. This enrichment, through a join, is what we will implement right away.

3 A few words on the principle of the star schema

3.1 Star schema 101

A dataset is rarely sufficient on its own. To give meaning and value to a statistic, it generally needs to be associated with additional knowledge, otherwise it remains detached. This kind of pairing, known as data enrichment, is a daily necessity for data scientists and one of the most common operations in data science.

It can happen at the same level as the original source (for example, linking, within a business database, customer information to data scattered across a sales file and a product file) or at different, generally more aggregated, conceptual levels to contextualize finer data (for example, associating individual travel times with those of the same age group or people living in the same municipality).

Here we will focus on the most favorable case, which is the situation where information allows for an exact match between two databases[2].

This practice stems from the fact that many information systems take the form of a star schema (@#fig-star-schema-en).

[2]Otherwise, we enter the world of fuzzy matching or probabilistic matching.

Fuzzy matching refers to situations where we no longer have an exact identifier to link two databases, but rather partially noisy information shared between two sources that can be used to establish the connection. For example, a product database might contain `Coca Cola 33CL` while another contains `Coca Cola can`, yet both names refer to the same product. Text distance techniques, for instance, can be used to link these observations together. This is typically what a search engine does.

Probabilistic matching is another type of approach. Here, observations in two databases are linked not on the basis of an identifier, but on the distance between a set of characteristics across the two databases. This technique is widely used in medical statistics or in public policy evaluation, based on *propensity score matching*.

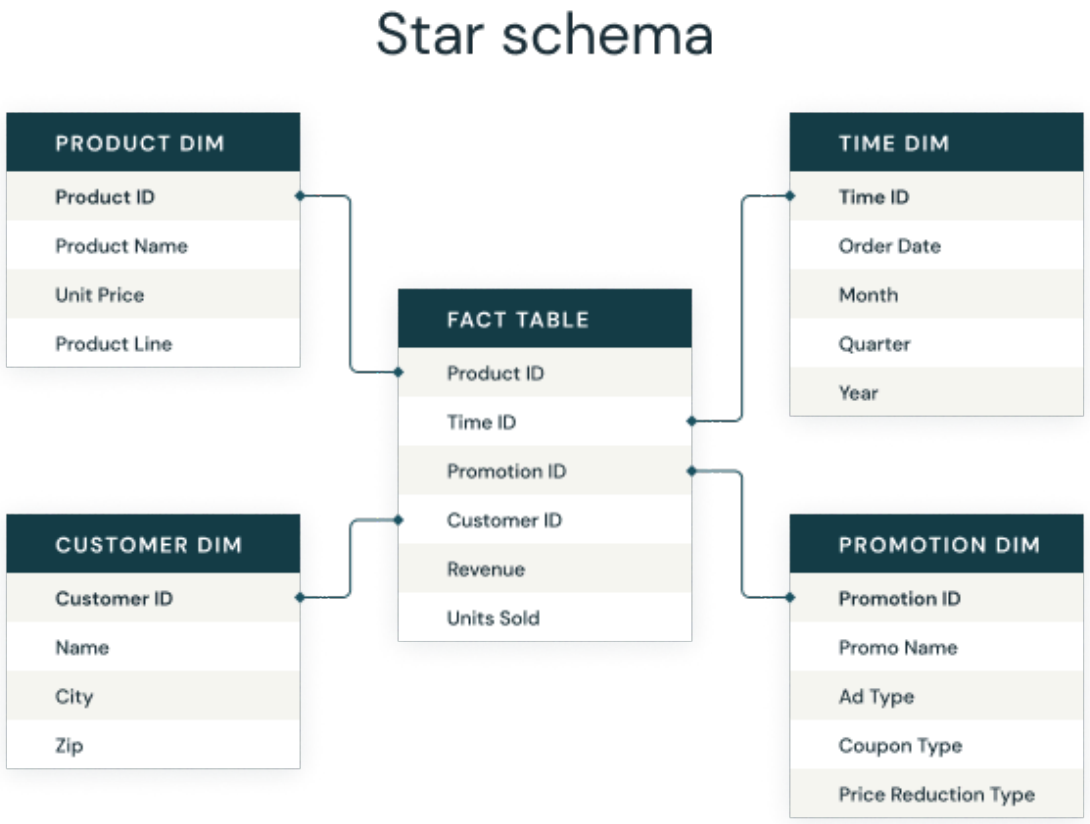


Figure 1: Illustration of the star schema (Source: Databricks)

This way of structuring information is closely tied to the relational table model from the 1980s. Today, more flexible data models exist, where information is stored in a *data lake* without any predefined structure. Nevertheless, the star schema model remains relevant because it makes it possible to organize information around statistical units of interest (the customer, the company, the transaction, etc.) while still allowing data to be compartmentalized.

Since the logic of the star schema historically comes from relational databases, it is natural that it is an approach intrinsically linked to the philosophy of SQL, even in the vocabulary. The term "data join" is often used, inherited from the SQL `JOIN` term, and the way to describe joins (left join, right join...) comes directly from the associated SQL instructions.

3.2 Joining keys

We refer to join keys as the variable(s) necessary for merging data. These are the variables common to both datasets. They do not need to have the same name, but they must share common values; otherwise, the intersection between these two datasets is the empty set.

We can manipulate two dimensions in the join (this will be clearer later with graphical examples):

- There are mainly three types of merges: left join, right join, or a combination of the two, depending on the type of pivot we want to implement.
- Then, there are two ways to merge the values once we have chosen a pivot: inner or outer join. In the first case, we only keep the observations where the join keys are present in both datasets; in the second, we keep all observations of the pivot key variables, even if the second dataset does not have such observations, resulting in missing values.

In the examples below, we will use the commune codes and departments as join keys. Using the department is not necessary since it is directly deduced from the commune code, but it helps illustrate the principle of joins on multiple variables. Note that the name of the commune is intentionally set

aside for joins, even though it is common information to both datasets. However, as it is a textual field, which may follow different formatting norms in the two datasets, it is not reliable for an exact join.

To illustrate the principle of the left or right pivot, we will create two identifier variables for the row in our left and right datasets. This will allow us to easily find rows present in one dataset but not in the other.

```
emissions = emissions.reset_index(names = ['id_left'])
filosofi = filosofi.reset_index(names = ['id_right'])
```

4 Implementation with `Pandas`

In `Pandas`, the most practical method to join datasets based on common characteristics is `merge`. Its main arguments allow for controlling the join behavior. We will explore them visually.

In our case, for constructing statistics on carbon emissions, the left base will be the `emissions` DataFrame, and the right base will be the `filosofi` DataFrame:

```
emissions.head(2)
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep
0	0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01
1	1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01

```
filosofi.head(2)
```

	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE	dep
0	0	01001	L'Abergement-Clémenciat	860	28270.0	NaN	01
1	1	01002	L'Abergement-de-Varey	270	28140.0	NaN	01

4.1 *Left join*

Let's start with the left join. As its name indicates, we will take the left variable as the pivot:

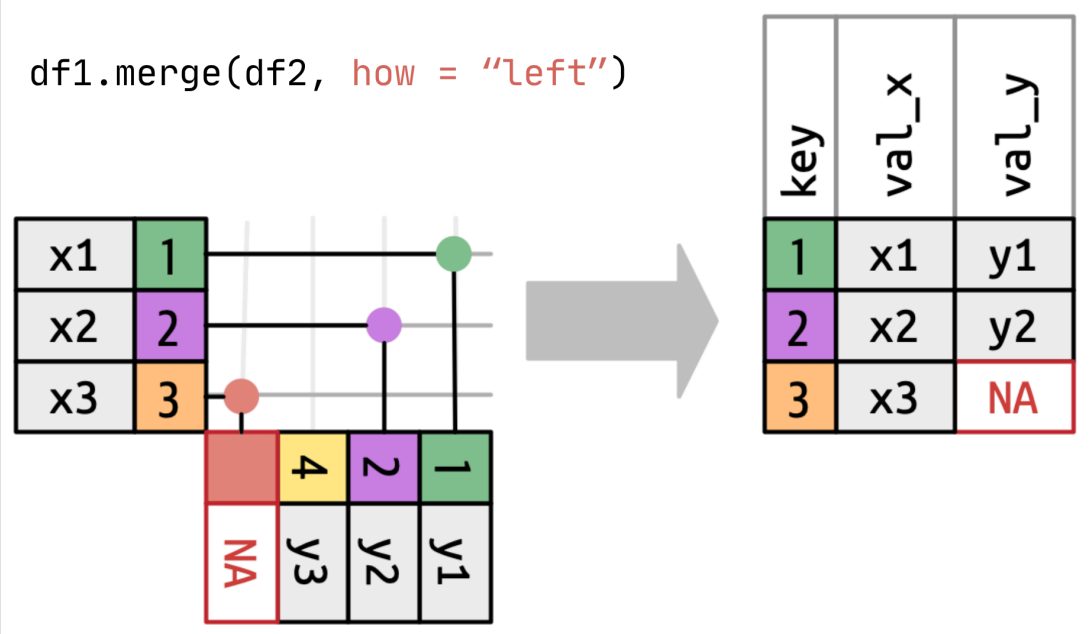


Figure 2: *Left join* illustration. Original image: H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1]

```
left_merged = emissions.merge(
  filosofi,
  left_on = ["INSEE commune", "dep"],
  right_on = ["CODGEO", "dep"],
  how = "left"
)
left_merged.head(3)
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE
0	0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01	0.0	01001	L'Abergement-Clémenciat	860.0	28270.0	NaN
1	1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01	1.0	01002	L'Abergement-de-Varey	270.0	28140.0	NaN
2	2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934	01	2.0	01004	Ambérieu-en-Bugey	15934.0	24210.0	18.0

It is recommended to always explicitly specify the join keys using the `left_on`, `right_on`, or `on` arguments if the variable names are common between the two datasets. If there are common variable names between the datasets that are not defined as join keys, they will not be used for the join but will be retained with a suffix that defaults to `_x` and `_y` (configurable using the `suffixes` argument).

The `Pandas` syntax is directly inspired by SQL, so we have a fairly transparent translation of the above instruction into SQL:

```
SELECT *
FROM emissions
LEFT JOIN filosofi
  ON emissions.`INSEE commune` = filosofi.CODGEO
  AND emissions.dep = filosofi.dep;
```

By performing a left join, we should, in principle, have as many rows as in the left dataset:

```
left_merged.shape[0] == emissions.shape[0]
```

```
True
```

Otherwise, it indicates that there is a duplicate key on the right. Thanks to our `id_right` variable, we can identify the commune codes on the right that do not exist on the left:

```
left_merged.loc[left_merged['id_right'].isna()].tail(3)
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE
35552	35552	95059	PIERREFITTE-SUR-SEINE	3.384360	293.997505	NaN	8462.785892	63344.922645	915.922961	2688.461993	22146.952752	24167.708776	14417.152611	93	NaN	NaN	NaN	NaN	NaN	NaN
35687	35687	95259	GADANCOURT	312.298700	NaN	NaN	142.113291	11.372909	NaN	NaN	32.440647	2060.981036	41.153991	95	NaN	NaN	NaN	NaN	NaN	NaN
35693	35693	95282	GOUZANGREZ	826.234401	NaN	NaN	53.146172	23.274790	2.354558	6.911213	55.638159	36.040818	84.222120	95	NaN	NaN	NaN	NaN	NaN	NaN

This is because the two datasets do not rely on the same version of the official geographical code. The Filosofi data, disseminated via the Melodi API, reflects the most recent municipal boundaries (2025), whereas the Ademe emissions data, although updated in 2025, has not yet caught up with some recent commune mergers. For example, the commune of Courcouronnes is still present as such in the emissions data (left base):

```
left_merged.loc[
  left_merged['id_right'].isna()
  & left_merged['Commune'].str.contains("COURCOURONNES", na=False)
]
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE
35348	35348	91182	COURCOURONNES	24.548795	103.360309	NaN	9623.065698	111.241872	1276.170296	3745.877636	9978.002088	57834.224552	10532.120761	91	NaN	NaN	NaN	NaN	NaN	NaN

Yet this commune has since merged with Evry to form the new commune of Évry-Courcouronnes, the only form under which it now appears in the Filosofi data (right base):

```
filosofi.loc[
  filosofi['LIBGEO']
  .str.lower()
  .str.contains("courcouronnes", na=False)
]
```

	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE	dep
34315	34315	91228	Évry-Courcouronnes	66919	21020.0	28.6	91

In a public statistics construction exercise, we could not afford this discrepancy in years.

4.2 *Right join*

The principle is the same, but this time it is the right base that is taken as the pivot:

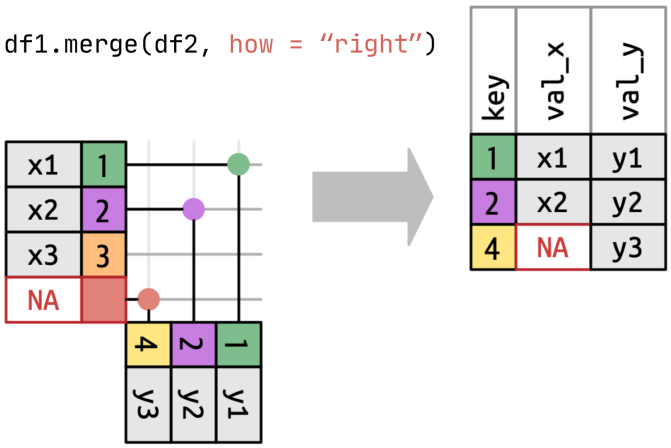


Figure 3: *Right join.* Original image from H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1]

```
right_merged = emissions.merge(
  filosofi,
```

```
  left_on = ["INSEE commune", "dep"],
  right_on = ["CODGEO", "dep"],
  how = "right"
)
right_merged.head(3)
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE
0	0.0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01	0	01001	L'Abergement-Clémenciat	860	28270.0	NaN
1	1.0	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01	1	01002	L'Abergement-de-Varey	270	28140.0	NaN
2	2.0	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10752.376934	01	2	01004	Ambérieu-en-Bugey	15934	24210.0	18.0

The equivalent instruction in SQL would be:

```
SELECT *
FROM filosofi
RIGHT JOIN emissions
  ON filosofi.CODGEO = emissions.`INSEE commune`
  AND filosofi.dep = emissions.dep;
```

We can, as before, check the consistency of the dimensions:

```
right_merged.shape[0] == filosofi.shape[0]
```

```
True
```

To check the number of rows in the Filosofi data that we do not have in our greenhouse gas emissions dataset, we can do:

```
right_merged['id_left'].isna().sum()
```

```
np.int64(120)
```

It's a small number. What are these observations?

```
right_merged.loc[
  right_merged['id_left'].isna(),
  filosofi.columns.tolist() + emissions.columns.tolist()
]
```

	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE	dep	id_left	INSEE commune	Commune	...	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep
4201	4201	12218	NaN	1540	24390.0	NaN	12	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	12
4338	4338	13055	Marseille	886040	22860.0	28.1	13	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	13
17537	17537	49126	NaN	17162	25770.0	7.0	49	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	49
27039	27039	69123	Lyon	519127	28290.0	18.0	69	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	69
29194	29194	75056	Paris	2103778	33650.0	16.8	75	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	75
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
34853	34853	97420	Sainte-Suzanne	25551	19560.0	34.9	97	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	97
34854	34854	97421	Salazie	7268	15180.0	53.0	97	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	97
34855	34855	97422	Le Tampon	82579	18470.0	38.1	97	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	97
34856	34856	97423	Les Trois-Bassins	7221	18980.0	35.0	97	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	97
34857	34857	97424	Cilaos	5040	15660.0	49.0	97	NaN	NaN	NaN	...	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	97

It is surprising to see that Paris, Lyon, and Marseille are present in the communal statistics dataset but not in the emissions dataset. To understand why, let's search in our emissions data for observations related to Marseille:

```
emissions.loc[
  emissions["Commune"]
  .str.upper()
  .str.contains('MARSEILLE-')
]
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep
4462	4462	13201	MARSEILLE-1ER-ARRONDISSEMENT	303.836964	850.796099	NaN	18749.902852	2572.980666	2366.477690	6946.201368	29755.542456	35324.129125	18535.087435	13
4463	4463	13202	MARSEILLE-2E-ARRONDISSEMENT	303.836964	83.384844	NaN	14248.781083	2451.230873	2366.477690	6946.201368	18629.702329	35324.129125	11530.097205	13
4464	4464	13203	MARSEILLE-3E-ARRONDISSEMENT	303.836964	18.238987	NaN	20705.839485	2627.535940	2366.477690	6946.201368	32283.391803	35324.129125	21695.331089	13
4465	4465	13204	MARSEILLE-4E-ARRONDISSEMENT	303.836964	84.063314	NaN	21525.141519	2649.386137	2366.477690	107849.501368	34633.530963	35324.129125	22961.055568	13
4466	4466	13205	MARSEILLE-5E-ARRONDISSEMENT	303.836964	NaN	NaN	21220.562849	2641.273890	2366.477690	6946.201368	46460.016400	35324.129125	22498.466001	13
4467	4467	13206	MARSEILLE-6E-ARRONDISSEMENT	303.836964	NaN	NaN	19817.462943	2602.050928	2366.477690	6946.201368	32135.350914	35324.129125	20219.051308	13
4468	4468	13207	MARSEILLE-7E-ARRONDISSEMENT	303.836964	NaN	NaN	17667.589643	2544.306929	2366.477690	6946.201368	26082.234669	35324.129125	17252.687971	13
4469	4469	13208	MARSEILLE-8E-ARRONDISSEMENT	303.836964	NaN	NaN	31033.598562	2905.706707	2366.477690	6946.201368	46457.720220	35324.129125	38019.611886	13
4470	4470	13209	MARSEILLE-9E-ARRONDISSEMENT	303.836964	206.323957	15168.902916	63507.303272	2866.252439	2366.477690	6946.201368	53791.270123	35324.129125	41201.045974	13
4471	4471	13210	MARSEILLE-10E-ARRONDISSEMENT	303.836964	NaN	NaN	23774.329289	2709.963639	2366.477690	6946.201368	39770.211148	35324.129125	27865.799133	13
4472	4472	13211	MARSEILLE-11E-ARRONDISSEMENT	303.836964	149.788396	NaN	24246.711719	2720.116339	2366.477690	11159.128368	41737.276404	35324.129125	32805.823144	13
4473	4473	13212	MARSEILLE-12E-ARRONDISSEMENT	303.836964	34.728224	NaN	25332.342243	2752.276194	2366.477690	6946.201368	44144.355145	35324.129125	28921.206190	13
4474	4474	13213	MARSEILLE-13E-ARRONDISSEMENT	303.836964	NaN	NaN	34480.339077	3001.079306	2366.477690	6946.201368	64717.674768	35324.129125	43333.716539	13
4475	4475	13214	MARSEILLE-14E-ARRONDISSEMENT	303.836964	250.117946	NaN	25588.294079	2758.463641	2366.477690	6946.201368	43612.783924	35324.129125	32142.611435	13
4476	4476	13215	MARSEILLE-15E-ARRONDISSEMENT	303.836964	320.003252	NaN	31565.381887	2919.403353	2366.477690	6946.201368	57187.164876	35324.129125	44748.956297	13
4477	4477	13216	MARSEILLE-16E-ARRONDISSEMENT	303.836964	518.266268	12.311853	12119.260477	2392.479039	2366.477690	20211.201368	12795.918860	35324.129125	10336.618838	13
23522	23522	60387	MARSEILLE-EN-BEAUVAISIS	330.191658	90.910961	NaN	1430.605647	191.355801	127.146118	373.205521	573.997868	2780.093648	692.439819	60

This is because the Ademe emissions dataset provides information on districts in the three largest cities, whereas the Insee dataset does not have this breakdown.

4.3 *Inner join*

This is the dataset where the keys are found at the intersection of the two tables.

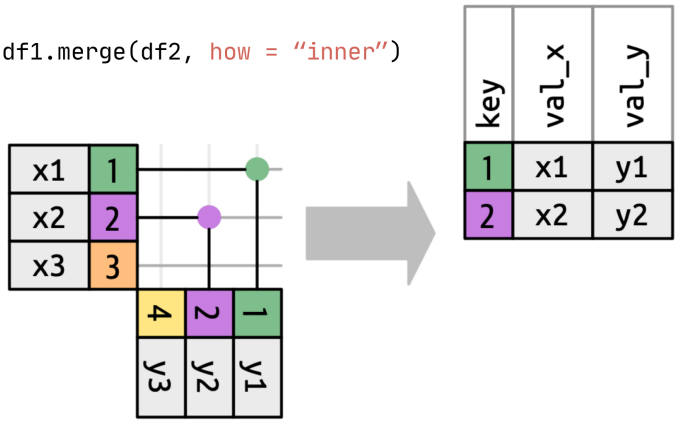


Figure 4: *Inner join*. Originale image: H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1]

```
inner_merged = emissions.merge(
  filosofi,
  left_on = ["INSEE commune", "dep"],
  right_on = ["CODGEO", "dep"],
  how = "inner"
)
inner_merged.head(3)
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE
0	0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01	0	01001	L'Abergement-Clémenciat	860	28270.0	NaN
1	1	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01	1	01002	L'Abergement-de-Varey	270	28140.0	NaN
2	2	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934	01	2	01004	Ambérieu-en-Bugey	15934	24210.0	18.0

In SQL, this would be:

```
SELECT *
FROM emissions
INNER JOIN filosofi
  ON emissions.`INSEE commune` = filosofi.CODGEO
  AND emissions.dep = filosofi.dep;
```

The number of rows in our dataset can be compared to the left and right datasets:

```
inner_merged.shape[0] == (
  left_merged.shape[0] - left_merged['id_right'].isna().sum()
)
```

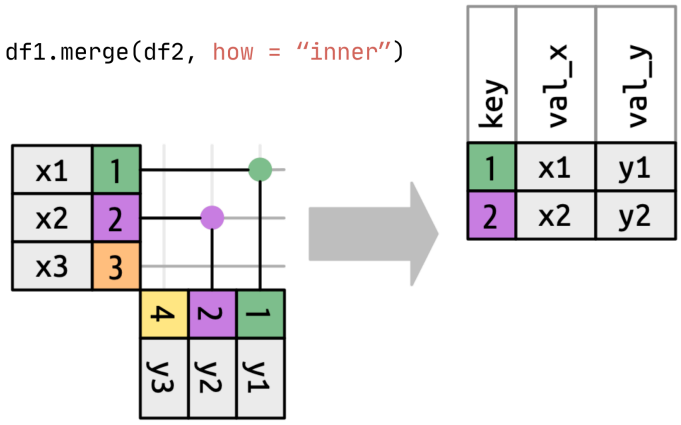
```
np.True_
```

```
inner_merged.shape[0] == (
  right_merged.shape[0] - right_merged['id_left'].isna().sum()
)
```

```
np.True_
```

4.4 *Full join*

The full join is a pivot to the left and then to the right for the information that was not found.




```
full_merged = emissions.merge(
  filosofi,
  left_on = ["INSEE commune", "dep"],
  right_on = ["CODGEO", "dep"],
  how = "outer"
)
full_merged.head(3)
```

	id_left	INSEE commune	Commune	Agriculture	Autres transports	Autres transports international	CO2 biomasse hors-total	Déchets	Energie	Industrie hors-énergie	Résidentiel	Routier	Tertiaire	dep	id_right	CODGEO	LIBGEO	POPULATION	NIVVIE_MEDIAN	TAUX_PAUVRETE
0	0.0	01001	L'ABERGEMENT-CLEMENCIAT	3711.425991	NaN	NaN	432.751835	101.430476	2.354558	6.911213	309.358195	793.156501	367.036172	01	0.0	01001	L'Abergement-Clémenciat	860.0	28270.0	NaN
1	1.0	01002	L'ABERGEMENT-DE-VAREY	475.330205	NaN	NaN	140.741660	140.675439	2.354558	6.911213	104.866444	348.997893	112.934207	01	1.0	01002	L'Abergement-de-Varey	270.0	28140.0	NaN
2	2.0	01004	AMBERIEU-EN-BUGEY	499.043526	212.577908	NaN	10313.446515	5314.314445	998.332482	2930.354461	16616.822534	15642.420313	10732.376934	01	2.0	01004	Ambérieu-en-Bugey	15934.0	24210.0	18.0

As usual, the translation to SQL is almost immediate:

```
SELECT *
FROM emissions
FULL OUTER JOIN filosofi
  ON emissions.`INSEE commune` = filosofi.CODGEO
  AND emissions.dep = filosofi.dep;
```

This time, we have a combination of our three initial datasets:

- The inner join;
- The left join on observations without the right key;
- The right join on observations without the left key;

```
(
  full_merged['id_left'].isna().sum() + full_merged['id_right'].isna().sum()
) == (
  left_merged['id_right'].isna().sum() + right_merged['id_left'].isna().sum()
)
```

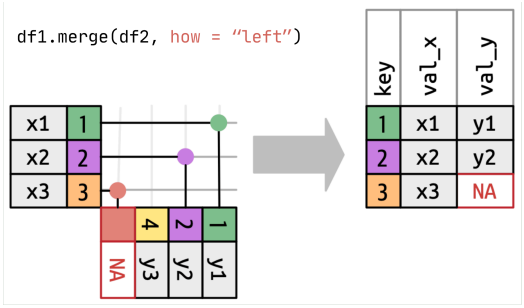
```
np.True_
```

The `id_left` and `id_right` columns were only useful to illustrate the joins above. We drop them from our datasets before moving on, so they do not interfere with later processing:

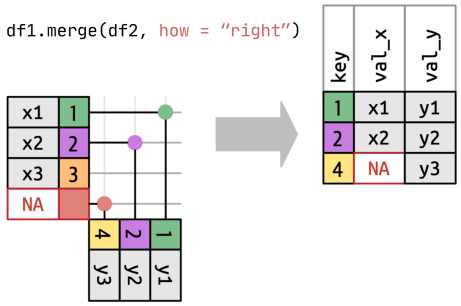
```
emissions = emissions.drop(columns = ['id_left'])
filosofi = filosofi.drop(columns = ['id_right'])
```

4.5 In summary

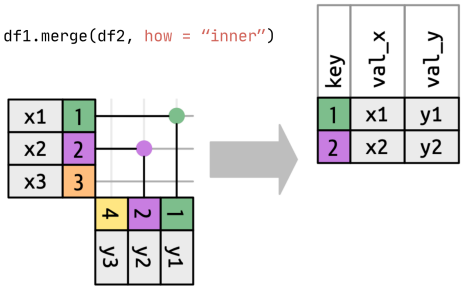
The four main types of join are the following



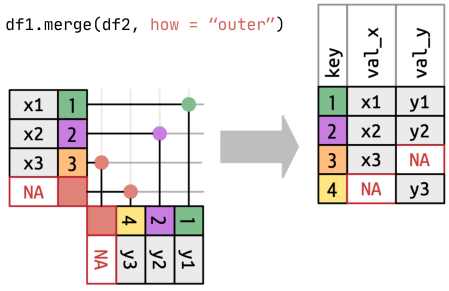
(a) *Left join* (source: H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1])



(b) *Right join* (source: H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1])



(c) *Inner join* (source: H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1])

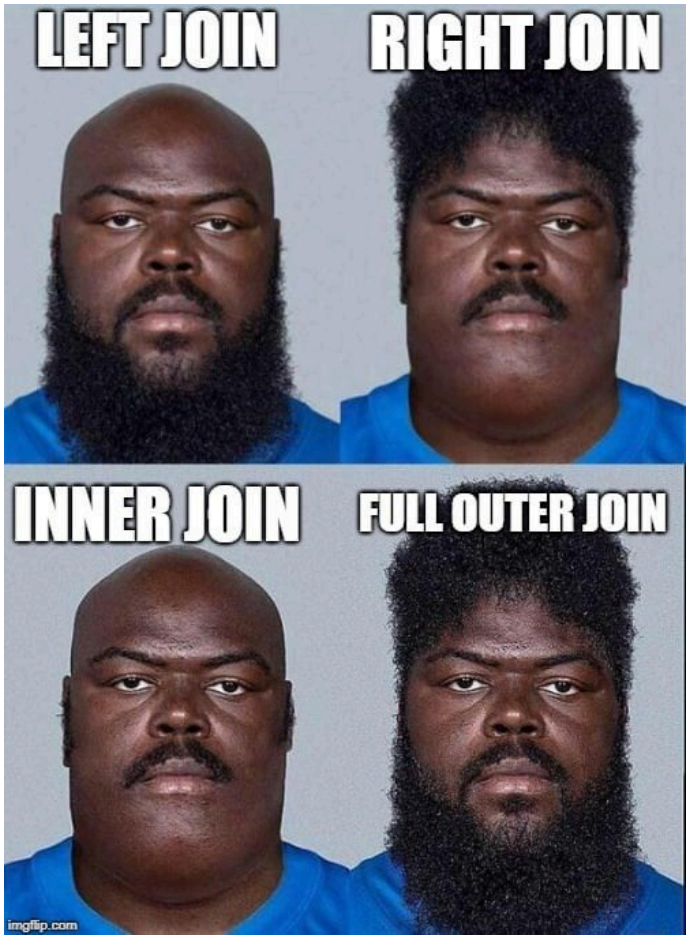


(d) *Full outer join* (source: H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund [1])

Main database join methods

Figure 5:

In a funny representation, this gives:



There are of course other types of joins, based on negation: *semi-join*, *anti-join*, etc. These methods are less frequently used but are useful when trying to determine the discrepancy between two datasets.

5 Examples of identifiers in French data

We have highlighted the value of having identifiers common to several databases rather than relying on noisy textual information. We will now discuss three identifiers that matter to users of French data:

- The official geographic code, a unique identifier for geographic data
- The NIR, or social security number, the French individual identifier
- The SIRENE code, a unique identifier for businesses operating in France.

5.1 The Official Geographic Code (COG): The identifier for geographic data

For geographic data, there are many identifiers depending on the study problem.

Among the main needs is the ability to match geographic data using a common administrative identifier. For example, associating two datasets at the municipal level.

For this, the reference identifier is the Insee code, derived from the Official Geographic Code (COG), which we have been using since the last chapter and will extensively use throughout the different chapters of this course. Given that the administrative geography is constantly evolving, the Insee

code database is a living base. The Insee website and APIs provide access to the post-war historical data for long-term geographic analysis.

Postal codes cannot be considered as an identifier: they can group several municipalities or, conversely, one municipality can have several postal codes. It is a system managed by La Poste that was not designed for statistical analysis.

To illustrate the problem, from the data provided by La Poste, we can see that postal code 11420 corresponds to 11 municipalities:

```
codes_postaux = pd.read_csv(
  "https://datanova.laposte.fr/data-fair/api/v1/datasets/laposte-hexasmal/raw",
  sep = ";", encoding = "latin1",
  dtype = {"Code_postal": "str", "#Code_commune_INSEE": "str"}
)
codes_postaux.loc[codes_postaux['Code_postal'] == "11420"]
```

	#Code_commune_INSEE	Nom_de_la_commune	Code_postal	Libellé_d_acheminement	Ligne_5
3921	11033	BELPECH	11420	BELPECH	NaN
3944	11057	CAHUZAC	11420	CAHUZAC	NaN
4080	11184	LAFAGE	11420	LAFAGE	NaN
4124	11226	MAYREVILLE	11420	MAYREVILLE	NaN
4134	11236	MOLANDIER	11420	MOLANDIER	NaN
4176	11277	PECHARIC ET LE PY	11420	PECHARIC ET LE PY	NaN
4177	11278	PECH LUNA	11420	PECH LUNA	NaN
4182	11283	PEYREFITTE SUR L HERS	11420	PEYREFITTE SUR L HERS	NaN
4189	11290	PLAIGNE	11420	PLAIGNE	NaN
4264	11365	ST SERNIN	11420	ST SERNIN	NaN
4317	11419	VILLAUTOU	11420	VILLAUTOU	NaN

Anticipating on the skills developed in the upcoming chapters, we can represent the problem cartographically by taking the example of the Aude department. The code to produce the map of commune codes is given as is, not developed, as it uses concepts and libraries that will be presented in the next chapter:

```
from cartiflette import carti_download

shp_communes = carti_download(
  values = ["11"],
  crs = 4326,
  borders = "COMMUNE",
  simplification=50,
  filter_by="DEPARTEMENT",
  source="EXPRESS-COG-CARTO-TERRITOIRE",
  year=2022)

codes_postaux11 = shp_communes.merge(
  codes_postaux,
  left_on = "INSEE_COM",
  right_on = "#Code_commune_INSEE"
)
codes_postaux11 = codes_postaux11.dissolve(by = "Code_postal")
```

Line 10
Downloading the official contours of Aude produced by IGN using the `cartiflette` library

Line 16
Joining using the commune code between the two data sources

Line 17
Aggregating the geometry at the postal code level

```
# Map
ax = shp_communes.plot(color='white', edgecolor='blue', linewidth = 0.5)
ax = codes_postaux11.plot(ax = ax, color='none', edgecolor='black')
ax.set_axis_off()
```

Line 1
Creating a map from our two layers

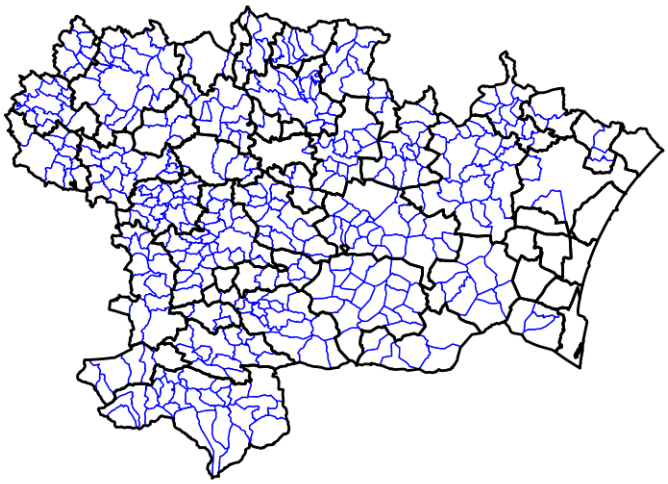


Figure 6: Geography of postal codes and municipalities in Aude (11)

5.1.a Sirene: the identifier in business data

To connect French business microdata, there is a unique identification number: the Siren number. It is an identification number in a legal business directory essential for all legal, fiscal, and other procedures. For companies that have multiple establishments—for example, in several cities—there is a derived identifier called the Siret: the 9 digits of the Siren number are followed by 5 establishment identification digits. Moreover, public administrations are also concerned with the Siren number: being involved in market operations (purchasing equipment, renting goods, etc.), they also have a Siren identifier. As they are registered in legal directories whose information is public, the Siren numbers and the associated company names are available in open data, for example, on annuaire-entreprises.data.gouv.fr/ for occasional searches, or on data.gouv.fr.

This Sirene database is a treasure trove of information, sometimes amusing, about French companies. For example, the site tif.hair/ cataloged the proportion of hair salons with puns in their names. When an entrepreneur declares the creation of a business, they receive a Siren number and an activity code (the APE code) related to the description of their business activity. This code allows the classification of a business activity in the French Classification of Activities (NAF), which will be used by Insee for the publication of sectoral statistics. In the case of hairdressers, the code in the NAF is 96.02A. From the open data available, it is possible, in a few lines of `Python`, to get the list of all hairdressers and then explore this data (the subject of the next optional exercise).

The following optional exercise suggests replicating, in a simplified manner, the survey done by tif.hair/ on puns in hair salon names. It allows practicing some text manipulation methods, ahead of the chapter dedicated to regular expressions.

Since the dataset of all companies is quite large (around 4GB in CSV after decompression), it is more practical to use a dataset in `Parquet` format, which is more optimized (more details on this format in the advanced chapter dedicated to it).

To read this type of file optimally, it is recommended to use the `DuckDB` library, which allows consuming only the necessary data instead of downloading the entire file to read only a part of it as would be the case with `Pandas` (see the last chapter of this part, section "Beyond `Pandas`"). The following SQL query translates into natural language as: *"From the `Parquet` file, I only want a few columns of the file for hairdressers (APE: 96.02A) whose business name (`denominationUsuelleEtablissement`) is provided"*:

```
import duckdb
coiffeurs = duckdb.sql("""
  SELECT
        siren, siret, dateDebut, enseigne1Etablissement, activitePrincipaleEtablissement,
denominationUsuelleEtablissement
  FROM
    read_parquet('https://minio.lab.sspcloud.fr/lgaliana/data/sirene2024.parquet')
  WHERE
    activitePrincipaleEtablissement == '96.02A'
    AND
    denominationUsuelleEtablissement IS NOT NULL
""")
coiffeurs = coiffeurs.df()
```

Line 12

Convert the DuckDB dataframe to a Pandas DataFrame.

```
coiffeurs.head(3)
```

	siren	siret	dateDebut	enseigne1Etablissement	activitePrincipaleEtablissement	denominationUsuelleEtablissement
0	024050379	02405037900023	2016-01-14	NaN	96.02A	SOPHA COIFFURE
1	024076481	02407648100019	2017-07-17	NaN	96.02A	SAF COIFFURE
2	047142872	04714287200036	2011-07-31	NaN	96.02A	JENNY

Optional Exercise: Punny Hairdressers

In this exercise, we will consider only the variable `denominationUsuelleEtablissement`.

1. In this dataset, `[ND]` is a code for missing value. Since `Python` has no reason to know this a priori and therefore didn't interpret these values as missing, use the `replace` method to replace `[ND]` with an empty text field. Also, recode missing values as an empty text field to avoid future errors related to the inability to apply certain text methods to missing values.
2. Search for all occurrences where the term `tif` appears, paying attention to the capitalization of the variable. Look at some observations.
3. Using this example, normalize the names of the salons by removing special characters and count the most frequent puns.

With question 2, we find a list of quite imaginative puns based on the term `tif`

```
"IMAGINA'TIF, DIMINUT'TIF, A L'INFINI'TIF, TIF'ANNICK, A TRAC TIF, BLEV HAIR TIF, CHRIS TIF,
OBJECTIF PRESTIGE, TIF'ALIZEA HAIRCUT, JS CREA TIF"
```

```
function underlineTif(x) {
  // Use a regular expression to find all occurrences of "tif"
  const modx = x.replace(/TIF/g, match ⇒
     `<span style="text-transform: capitalize; border-bottom: solid 2px blue; margin-bottom:
-2px;">${match}</span>`
  );

  console.log(modx)

  // Return the modified string wrapped in an HTML template
  return html`${modx}`;
}
```

In a more interactive form, here's a list of all the hairdressers who have the word `tif` in the name of their registered business in the official data:

Of course, to go further, it would be better to normalize the data more thoroughly, check that the information sought is not spread across multiple columns, and conduct visual inspections to detect hidden puns. But already, in just a few minutes, we have partial statistics on the phenomenon of punny hairdressers.

5.1.b The social security number and the issue of individual identifiers' confidentiality

For individuals, there exists a unique identifier that allows linking them across different data sources: the NIR, also known as the INSEE number or social security number. This number is necessary for the administration to manage social benefits (health, retirement, family…). Beyond this function, which can be useful daily, this number is a unique individual identifier in the National Register of Physical Persons (RNIPP).

This identifier is mainly present in management databases related to payroll, social benefits, etc. However, unlike the Sirene number, it contains several sensitive pieces of information and is inherently linked to the sensitive issue of social security rights.



Figure 7: Social security number (adapted from French version)

To address this problem, the non-significant statistical code (CSNS) or hashed NIR, a non-identifying anonymous individual identifier, was recently implemented. The goal of this anonymized identifier is to reduce the dissemination of personal information that, although allowing civil servants and researchers to deterministically link numerous databases, provided analysts with non-essential information about the individuals in question. Y.-L. Bénichou, L. Espinasse, and S. Gilles [2] describe in more detail how French official statistics organizations can use this unique non-significant code to link various data sources with one another.

5.2 Why do we need a commune code when we already have its name?

This exercise will take a step back to understand why we assumed above that the commune code was the key for joining data.

Tip 1: Verification of Join Keys

Let's start by checking the dimensions of the `DataFrames` and the structure of some key variables. In this case, the fundamental variables for linking our data are the communal variables. Here, we have two geographical variables: a commune code and a commune name.

1. Check the dimensions of the `DataFrames`.
2. Identify in `filosofi` the commune names that correspond to multiple commune codes and select their codes. In other words, identify the `LIBGEO` where there are duplicate `CODGEO` and store them in a vector `x` (tip: be careful with the index of `x`).

We temporarily focus on observations where the label involves more than two different commune codes.

- *Question 3.* Look at these observations in `filosofi`.
- *Question 4.* To get a better view, reorder the obtained dataset alphabetically.
- *Question 5.* Determine the average size (variable number of people: `POPULATION`) and some descriptive statistics of this data. Compare it to the same statistics on the data where labels and commune codes coincide.
- *Question 6.* Check the major cities (more than 100,000 people) for the proportion of cities where the same name is associated with different commune codes.
- *Question 7.* Check in `filosofi` the cities where the label is equal to Montreuil. Also, check those that contain the term *'Saint-Denis'*.

This small exercise reassures us as the duplicated labels are actually the same commune names but in different departments. So, these are not duplicated observations. We can thus rely on the commune codes, which are unique.

5.3 Associating different sources to compute carbon footprints

Tip 2: Calculate the carbon footprint per capita

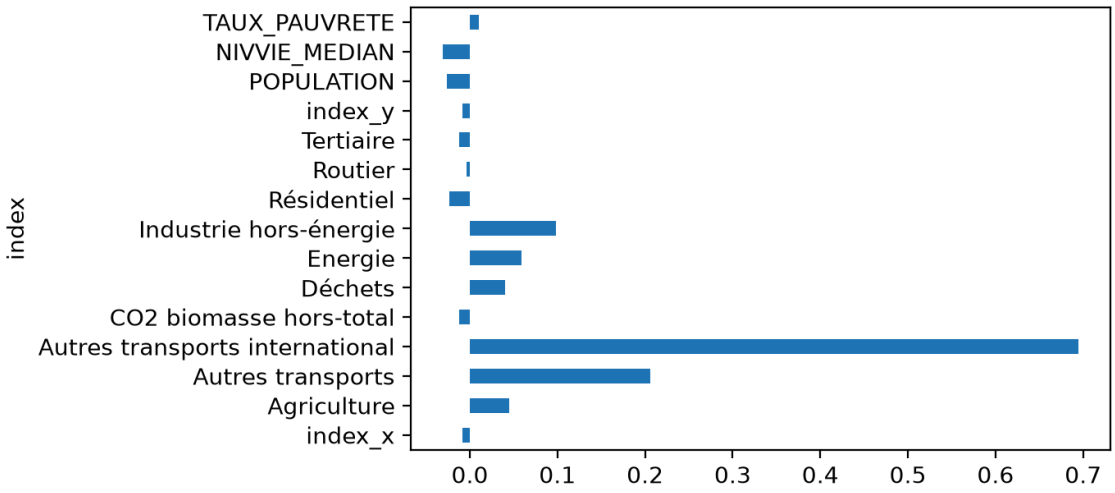
First, we will calculate the carbon footprint of each commune.

1. Create a variable `emissions` that corresponds to the total emissions of a commune.
2. Perform a left join between the emissions data and the framing data[^notebiais].
3. Calculate the carbon footprint (total emissions / population).

At this stage, we might want to move towards modeling to try to explain the determinants of the carbon footprint based on communal variables. However, for an inferential approach to be relevant, it is necessary to check some descriptive statistics beforehand.

4. Generate a histogram of the carbon footprint per commune in level and log.
5. Look at the correlation between the framing variables and the carbon footprint. Do some variables seem to potentially influence the carbon footprint?

At the end of question 5, the correlation graph is as follows:



With a better understanding of our data, gained through this association of sources, we get closer to inferential statistics. In the next chapter, we will continue to use `emissions` and `filosofi`, enriched by one another, to build group descriptive statistics and restructure our data.

Informations additionnelles

i Python environment

This site was built automatically through a `Github` action using the `Quarto` reproducible publishing software (version 1.10.18).

The environment used to obtain the results is reproducible via uv. The `pyproject.toml` file used to build this environment is available on the `linogaliana/python-datascientist` repository

```
[project]
name = "python-datascientist"
version = "0.1.0"
description = "Source code for Lino Galiana's Python for data science course"
readme = "README.md"
requires-python = "≥3.13,<3.14"
dependencies = [
    "altair≥6.0.0",
    "cartiflette≥0.2",
    "contextily==1.6.2",
    "duckdb≥0.10.1",
    "folium≥0.19.6",
    "gdal==3.11.4",
    "graphviz==0.20.3",
    "great-tables≥0.12.0",
    "gt-extras≥0.0.8",
    "ipykernel≥6.29.5",
    "jupyter≥1.1.1",
    "jupyter-cache≥1.0.0",
    "kaleido≥0.2.1",
    "langchain-community≥0.3.27",
    "loguru==0.7.3",
    "markdown≥3.8",
    "nbclient≥0.10.0",
    "nbformat≥5.10.4",
```

```
    "nltk>=3.9.1",
    "openpyxl>=3.1.5",
    "pandas>=3.0",
    "pip>=25.1.1",
    "plotly>=6.1.2",
    "plotnine>=0.15",
    "polars>=1.8.2",
    "py7zr>=1.1.3",
    "pyarrow>=17.0.0",
    "pynsee>=0.1.8",
    "python-dotenv>=1.0.1",
    "python-frontmatter>=1.1.0",
    "pywaffle>=1.1.1",
    "requests>=2.32.3",
    "scikit-image>=0.24.0",
    "scikit-learn>=1.8.0",
    "scipy>=1.13.0",
    "seaborn>=0.13.2",
    "selenium<4.39.0",
    "spacy>=3.8.4",
    "webdriver-manager>=4.0.2",
    "wordcloud==1.9.3",
]

[tool.uv.sources]
gdal = [
  { index = "gdal-wheels", marker = "sys_platform == 'linux'" },
  { index = "geospatial_wheels", marker = "sys_platform == 'win32'" },
]

[[tool.uv.index]]
name = "geospatial_wheels"
url = "https://nathanjmcdougall.github.io/geospatial-wheels-index/"
explicit = true

[[tool.uv.index]]
name = "gdal-wheels"
url = "https://gitlab.com/api/v4/projects/61637378/packages/pypi/simple"
explicit = true

[dependency-groups]
dev = [
    "nb-clean>=4.0.1",
]
```

To use exactly the same environment (version of `Python` and *packages*), please refer to the documentation for `uv`.

Bibliography

[1] H. Wickham, M. Çetinkaya-Rundel, and G. Grolemund, *R for data science*. " O'Reilly Media, Inc.", 2023.

[2] Y.-L. Bénichou, L. Espinasse, and S. Gilles, “Le Code statistique non signifiant (CSNS) : un service pour faciliter les appariements de fichiers,” *Courrier des Statistiques*, no. 9, pp. 64–85, Jun. 2023, [Online]. Available: https://insee.hal.science/hal-05295572